DO NOW — RECALL

Floating point → denary

Name: _____ Class: _____ Date: _____

1. Represent 6.5 in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

2. Convert $6D_{16}$ to eight-bit binary and denary.

3. Convert 11110110 from 8-bit two’s-complement binary to denary. Show signed place values.

4. Apply AND 00001111 to 10110110 . Give the result in binary and hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

01101000 0011 6.5 = 110.1$_2$ = 0.1101$_2$ × 2^3 . Mantissa 01101000 ; exponent 0011 .

2. Model answer

01101101$_2$ and 109 . 6 → 0110 , D → 1101 . In denary: (6 × 16) + 13 = 109 .

3. Model answer

−10 −128 + 64 + 32 + 16 + 4 + 2 = −10 .

4. Model answer

00000110$_2$ = 06$_{16}$ AND with 00001111 clears the upper nibble and preserves the lower nibble 0110 .

By the end, you can…

• evaluate the signed mantissa and exponent separately;

• convert floating-point representations with positive or negative values and exponents to denary;

• recognise normalisation and explain precision versus range.

Vocabulary — know these terms

Recall the term — Mantissa

Recall the term — Exponent

Recall the term — Normalised

Complete the statement — Precision

Complete the statement — Range

Recall the representation — Two’s complement

NEW LEARNING

Read, annotate and explain

Split → evaluate → scale

8-bit mantissa: −1 | ½ ¼ ⅛ 1⁄16 1⁄32 1⁄64 1⁄128
4-bit exponent: −8 4 2 1

Value = mantissa × 2^{E}

Two signs, two jobs: the mantissa controls the sign of the value ; the exponent controls its scale .

Split the fields

Separate the bit pattern into the mantissa and the exponent at the stated field boundary.

Evaluate the mantissa

Add the place values for the bits that are 1.

Convert the exponent to denary

Read the exponent as a signed two’s-complement integer .

Apply the scale

A positive exponent multiplies by a power of two. A negative exponent divides by a power of two.

Check the representation

For a non-zero normalised mantissa:

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Convert this floating-point representation to denary

Evaluate the mantissa

```
01011000
= ½ + ⅛ + 1/16
= 0.6875
```

Read the mantissa as a signed binary fraction.

Convert the exponent to denary

```
0011 = 3
```

The exponent is a signed two’s-complement integer.

Apply the scale

```
0.6875 × 2³
= 0.6875 × 8
= 5.5
```


WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the answer

Guided 2 — CHECK

Build the answer

Guided 3 — CHECK

Build the answer

Guided 4 — CHECK

Build the answer
Shift the mantissa left twice and reduce the exponent by 2.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A negative exponent means the floating-point value must be negative.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Convert this floating-point representation to denary

Convert this floating-point representation to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 2 — TRY: Convert this floating-point representation to denary

Convert this floating-point representation to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 3 — TRY: Convert this floating-point representation to denary

Convert this floating-point representation to denary. Use an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

Guided 4 — TRY: Convert to denary, then write in normalised floating-point form

Convert the value to denary, then write the same value in normalised floating-point form using an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Convert 0100 1110 to denary
The mantissa is four bits and the exponent is four bits. Both use two’s complement. Show your working.
Paraphrased OCR June 2023 Q3(d); original values and marks retained. Source .

Question 2

Convert 1010 1110 to denary
Use a four-bit mantissa and four-bit exponent. Show each field value, the scaling operation and the answer.
Paraphrased OCR June 2022 Q3(b); original values and marks retained. Source .

Question 3

Which representations are normalised?
Each representation uses a six-bit two’s-complement mantissa followed by a three-bit two’s-complement exponent. State which representations are normalised by checking the mantissa’s leading bits.
CODE:
a) 010101 100
b) 110101 111
c) 011010 010
d) 101010 110
Paraphrased OCR June 2022 Q3(c); original marks retained. Source .

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 4

Convert both floating-point representations to denary
(a) 01110000 0010 (b) 10010000 1111
Each representation uses an eight-bit two’s-complement mantissa and a four-bit two’s-complement exponent. Convert both to denary and show the mantissa value, exponent value and final denary result.

Question 5

Precision and range
What improves if more bits are allocated to (a) the mantissa and (b) the exponent?
Then explain the trade-off if the total number of bits is fixed.
Think about what mantissa bits control and what exponent bits control.
The first two prompts paraphrase OCR June 2023 Q3(e). The trade-off prompt is an original classroom extension. Source .